PROPER BASE CHANGE FOR TORSION ETALE SHEAVES ON ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- Scheme proper base change, algebraic-space etale stalks, filtered colimits of cohomology, the acyclic-embedding criterion and Chow covers.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original lemma 04K2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5958–5958. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 089J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10499–10499. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0DES: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16812–16812. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0DDW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16781–16781. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0A0C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12630–12630. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0A3U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12752–12752. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original section 095S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11595–11595. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Date: stacks-project:spaces-more-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 07U6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9961–9961. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 015E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1505–1505. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Lemma 0.2. *Let (A, I) be a henselian pair. Let X be an algebraic space over A such that the structure morphism $f : X \to \mathrm{Spec}(A)$ is proper. Let $i : X_0 \to X$ be the inclusion of $X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I)$. For any sheaf $\mathcal{F}$ on $X_{étale}$ we have $\Gamma(X, \mathcal{F}) = \Gamma(X_0, i^{-1}\mathcal{F})$.*

Proof. Choose a surjective proper morphism $Y \to X$ where Y is a scheme, see Cohomology of Spaces, Lemma 089J. Consider the diagram

$$\begin{array}{ccccc} \Gamma(X_0, \mathcal{F}_0) & \longrightarrow & \Gamma(Y_0, \mathcal{G}_0) & \rightrightarrows & \Gamma((Y \times_X Y)_0, \mathcal{H}_0) \\ \uparrow & & \uparrow & & \uparrow \\ \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(Y, \mathcal{G}) & \rightrightarrows & \Gamma(Y \times_X Y, \mathcal{H}) \end{array}$$

Here $\mathcal{G}$, resp. $\mathcal{H}$ is the pullbackf or $\mathcal{F}$ to Y, resp. $Y \times_X Y$ and the index 0 indicates base change to $\mathrm{Spec}(A/I)$. By the case of schemes (Étale Cohomology, Lemma 0A0C) we see that the middle and right vertical arrows are bijective. By Lemma 0.1 it follows that the left one is too. □

Lemma 0.3. *Let A be a henselian local ring. Let X be an algebraic space over A such that $f : X \to \mathrm{Spec}(A)$ is a proper morphism. Let $X_0 \subset X$ be the fibre of f over the closed point. For any sheaf $\mathcal{F}$ on $X_{étale}$ we have $\Gamma(X, \mathcal{F}) = \Gamma(X_0, \mathcal{F}|_{X_0})$.*

Proof. This is a special case of Lemma 0.2. □

Lemma 0.4. *Let S be a scheme. Let $f : X \to Y$ and $g : Y' \to Y$ be a morphisms of algebraic spaces over S. Assume f is proper. Set $X' = Y' \times_Y X$ with projections $f' : X' \to Y'$ and $g' : X' \to X$. Let $\mathcal{F}$ be any sheaf on $X_{étale}$. Then $g^{-1}f_*\mathcal{F} = f'_*(g')^{-1}\mathcal{F}$.*

Proof. The question is étale local on Y'. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme V' and a surjective étale morphism $V' \to V \times_Y Y'$. Then we may replace Y' by V' and Y by V. Hence we may assume Y and Y' are schemes. Then we may work Zariski locally on Y and Y' and hence we may assume Y and Y' are affine schemes.

Assume Y and Y' are affine schemes. Choose a surjective proper morphism $h_1 : X_1 \to X$ where X_1 is a scheme, see Cohomology of Spaces, Lemma 089J. Set $X_2 = X_1 \times_X X_1$ and denote $h_2 : X_2 \to X$ the structure morphism. Observe this is a scheme. By the case of schemes (Étale Cohomology, Lemma 0A3U) we know the lemma is true for the cartesian diagrams

$$\begin{array}{ccc} X_1' & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array} \quad \text{and} \quad \begin{array}{ccc} X_2' & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

and the sheaves $\mathcal{F}_i = (X_i \to X)^{-1}\mathcal{F}$. By Lemma 0.1 we have an exact sequence $0 \to \mathcal{F} \to h_{1,*}\mathcal{F}_1 \to h_{2,*}\mathcal{F}_2$ and similarly for $(g')^{-1}\mathcal{F}$ because $X_2' = X_1' \times_{X'} X_1'$. Hence we conlude that the lemma is true (some details omitted). □

Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $\overline{x} : \mathrm{Spec}(k) \to S$ be a geometric point. The fibre of f at $\overline{x}$ is the algebraic space $Y_{\overline{x}} = \mathrm{Spec}(k) \times_{\overline{x},X} Y$ over $\mathrm{Spec}(k)$. If $\mathcal{F}$ is a sheaf on $Y_{étale}$, then denote $\mathcal{F}_{\overline{x}} = p^{-1}\mathcal{F}$ the pullback of $\mathcal{F}$ to $(Y_{\overline{x}})_{étale}$. Here $p : Y_{\overline{x}} \to Y$ is the projection. In the following we will consider the set $\Gamma(Y_{\overline{x}}, \mathcal{F}_{\overline{x}})$.

Lemma 0.5. *Let S be a scheme. Let $f : Y \to X$ be a proper morphism of algebraic spaces over S. Let $\overline{x} \to X$ be a geometric point. For any sheaf $\mathcal{F}$ on $Y_{étale}$ the canonical map*

$$(f_*\mathcal{F})_{\overline{x}} \longrightarrow \Gamma(Y_{\overline{x}}, \mathcal{F}_{\overline{x}})$$

is bijective.

Proof. This is a special case of Lemma 0.4. □

Theorem 0.6. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow[g']{} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian square of algebraic spaces over S. Assume f is proper. Let $\mathcal{F}$ be an abelian torsion sheaf on $X_{étale}$. Then the base change map

$$g^{-1}Rf_*\mathcal{F} \longrightarrow Rf'_*(g')^{-1}\mathcal{F}$$

is an isomorphism.

Proof. This proof repeats a few of the arguments given in the proof of the proper base change theorem for schemes. See Étale Cohomology, Section 095S for more details.

The statement is étale local on Y' and Y, hence we may assume both Y and Y' are affine schemes. Observe that this in particular proves the theorem in case f is representable (we will use this below).

For every $n \geq 1$ let $\mathcal{F}[n]$ be the subsheaf of sections of $\mathcal{F}$ annihilated by n. Then $\mathcal{F} = \mathrm{colim}\, \mathcal{F}[n]$. By Cohomology of Spaces, Lemma 07U6 the functors $g^{-1}R^pf_*$ and

$R^p f'_*(g')^{-1}$ commute with filtered colimits. Hence it suffices to prove the theorem if $\mathcal{F}$ is killed by n.

Let $\mathcal{F} \to \mathcal{I}^\bullet$ be a resolution by injective sheaves of $\mathbf{Z}/n\mathbf{Z}$-modules. Observe that $g^{-1}f_*\mathcal{I}^\bullet = f'_*(g')^{-1}\mathcal{I}^\bullet$ by Lemma 0.4. Applying Leray's acyclicity lemma (Derived Categories, Lemma 015E) we conclude it suffices to prove $R^p f'_*(g')^{-1}\mathcal{I}^m = 0$ for $p > 0$ and $m \in \mathbf{Z}$.

Choose a surjective proper morphism $h : Z \to X$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. Choose an injective map $h^{-1}\mathcal{I}^m \to \mathcal{J}$ where $\mathcal{J}$ is an injective sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules on $Z_{étale}$. Since h is surjective the map $\mathcal{I}^m \to h_*\mathcal{J}$ is injective (see Lemma 0.1). Since $\mathcal{I}^m$ is injective we see that $\mathcal{I}^m$ is a direct summand of $h_*\mathcal{J}$. Thus it suffices to prove the desired vanishing for $h_*\mathcal{J}$.

Denote h' the base change by g and denote $g'' : Z' \to Z$ the projection. There is a spectral sequence

$$E_2^{p,q} = R^p f'_* R^q h'_*(g'')^{-1}\mathcal{J}$$

converging to $R^{p+q}(f' \circ h')_*(g'')^{-1}\mathcal{J}$. Since h and $f \circ h$ are representable (by schemes) we know the result we want holds for them. Thus in the spectral sequence we see that $E_2^{p,q} = 0$ for $q > 0$ and $R^{p+q}(f' \circ h')_*(g'')^{-1}\mathcal{J} = 0$ for $p + q > 0$. It follows that $E_2^{p,0} = 0$ for $p > 0$. Now

$$E_2^{p,0} = R^p f'_* h'_*(g'')^{-1}\mathcal{J} = R^p f'_*(g')^{-1}h_*\mathcal{J}$$

by Lemma 0.4. This finishes the proof. □